

Cylinder Force Derivation

Wall Hydraulics — static moment balance about the hinge. This is the relation implemented by `compute_F_piston` in `wall.py`, validated against the ± 14.6 N/kg hand check.

1. Setup

The wall is a rigid body hinged at the origin, tilted to angle θ from the floor ($\theta = 0$ flat, $\theta = 90^\circ$ fully up). The motion is quasi-static: at every angle the cylinder holds the wall in equilibrium, so the net torque about the hinge is zero.

Points fixed to the wall, measured from the hinge (body frame):

- Piston attachment: (b, d) — distance b up the wall, offset d off it.
- Center of gravity: (x_cg, z_cg).
- Cylinder base: (-a, f) — bolted to the floor, so it does NOT rotate.

2. Rotate the wall's points into the world (by θ)

Applying the rotation matrix $[[\cos, -\sin], [\sin, \cos]]$:

$$\begin{aligned} \text{attachment} &= (b \cdot \cos\theta - d \cdot \sin\theta, b \cdot \sin\theta + d \cdot \cos\theta) \\ \text{cg} &= (x_{\text{cg}} \cdot \cos\theta - z_{\text{cg}} \cdot \sin\theta, x_{\text{cg}} \cdot \sin\theta + z_{\text{cg}} \cdot \cos\theta) \end{aligned}$$

3. Torque 1 — gravity

Weight (0, $-m \cdot g$) acts at the cg. Its torque about the hinge is the weight times the HORIZONTAL moment arm:

$$\tau_{\text{grav}} = -m \cdot g \cdot x_{\text{cg_world}} = -m \cdot g \cdot r_{\text{cg}} \cdot \cos\alpha$$

where $r_{\text{cg}} = \sqrt{x_{\text{cg}}^2 + z_{\text{cg}}^2}$ is constant and $\alpha = \text{atan2}(z_{\text{cg_world}}, x_{\text{cg_world}})$ is the angle of the cg vector. The product $r_{\text{cg}} \cdot \cos\alpha$ is just the horizontal distance from hinge to cg.

4. Torque 2 — the piston

The cylinder force F acts along the line from its base (-a, f) to the attachment point. Its torque about the hinge is force times the PERPENDICULAR distance from the hinge to that line of action:

$$\tau_{\text{piston}} = F \cdot r_{\text{att}} \cdot \sin(\beta - \varphi) \quad \leftarrow \text{the lever arm}$$

with three geometric angles:

- $r_{\text{att}} = \sqrt{b^2 + d^2}$ — distance hinge to attachment (constant on the wall).
- $\beta = \text{atan2}(z_{\text{att}}, x_{\text{att}})$ — direction of the attachment radius from the hinge.
- $\varphi = \text{atan2}(z_{\text{att}} - f, x_{\text{att}} + a)$ — direction of the cylinder line (base $\rightarrow$ attachment).

$(\beta - \varphi)$ is the angle between the attachment radius and the cylinder line, so $r_{\text{att}} \cdot \sin(\beta - \varphi)$ is the effective lever arm — the mechanical-advantage term.

5. Equilibrium — solve for F

Set $\tau_{\text{grav}} + \tau_{\text{piston}} = 0$ and solve:

$$F = \frac{-m \cdot g \cdot r_{\text{cg}} \cdot \cos\alpha}{r_{\text{att}} \cdot \sin(\beta - \varphi)}$$

6. What each piece tells you

- Numerator ($-m \cdot g \cdot r_{cg} \cdot \cos\alpha$): the gravity load. Largest when the cg is farthest out horizontally ($\theta = 0$, wall flat); vanishes as the cg swings over the hinge — which is why demand peaks at the bottom.
- Denominator ($r_{att} \cdot \sin(\beta - \phi)$): the lever arm / mechanical advantage.
- Sign of F : the push/pull convention — $F > 0$ = pull (retract), $F < 0$ = push (extend).
- Singularity: when $\sin(\beta - \phi) = 0$ ($\beta = \phi$) the cylinder line passes through the hinge, the lever arm is zero, and $F \rightarrow \infty$. That is the over-center pole the app caps at 50 N/kg.

Per unit mass, set $m = 1$ and F is in N/kg, exactly as the app reports.